

Dev Patel

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Education

University of Florida

Master of Science in Computer Science; GPA: 3.8 / 4.0

Aug 2022 - May 2024

Gainesville, FL, United States

Dharmsinh Desai University

Bachelor of Engineering in Computer Engineering; CGPA: 8.9 / 10

Aug 2018 - May 2022

Nadiad, GJ, India

Skills

- **Languages:** Python, SQL, TypeScript/JavaScript, Java | (Familiar: Go, Rust, C++, C#, Dart)
- **Data Engineering & Analytics:** ETL pipelines, data modeling, data quality auditing, pandas, NumPy, time-series forecasting, Bayesian hyperparameter optimization, query optimization
- **Visualization & Full-Stack:** Matplotlib, SHAP/Grad-CAM explainability, custom React/Next.js dashboards, FastAPI, Flask, Node.js, Express, REST APIs, SSE, Flutter
- **Databases/Search:** PostgreSQL, MySQL, Redis, MongoDB, Pinecone, vector search, Redis queues
- **Cloud/DevOps:** AWS (EC2, ALB, RDS, S3), GCP (Cloud Run, AlloyDB), Azure (Functions, AKS), Docker, Kubernetes, CI/CD with GitHub Actions, automated testing (Pytest), Nginx, Sentry monitoring, Git, Jira, Linux
- **GenAI/ML:** PyTorch, HuggingFace Diffusers, LLMs, OpenAI-compatible APIs, MCP, prompt engineering, diffusion models

Experience

- **Research Assistant (SMILE Lab)** | Data & ML Platform Engineering, University of Florida Dec 2024 - Present
 - Built **CROWN** from scratch and owned it end to end: a full-stack **Python (FastAPI) + React (Next.js)** platform for whole-head **MRI segmentation** and TES simulation, supporting Nifti uploads, **6 segmentation models**, Dockerized inference, and downloadable outputs for researchers.
 - Engineered cross-cohort **ETL and data-harmonization pipelines** across **OASIS/ADNI/NACC** datasets with automated **data quality checks**, logging, monitoring, fairness auditing, and Bayesian Hyperparameter Optimization, improving AD-prediction AUC by **12%**.
 - Distilled complex model behavior into clear visual explanations (**Grad-CAM, SHAP**) for clinicians & non-technical stakeholders for improving clinician-rated interpretability of **ResNet, AlexNet, Xception, & Inception** models by **25%**.
 - Built reliable data infrastructure for multi-model segmentation workflows using **Redis-backed GPU job scheduling** and real-time **SSE progress dashboards**, enabling researchers to monitor inference jobs and access 3D outputs.
 - Published and documented **crown-cli**, a Python CLI package standardizing reproducible segmentation workflows; drove adoption among lab researchers and automated batch inference.
 - Improved **text-to-image** generation stability and batch throughput by **~25%** via **HuggingFace Diffusers** pipelines with SLERP latent interpolation and a fully assembled UnCLIP pipeline.
- **Software Developer** | *Indian Institute of Technology Bombay, Mumbai, India* Dec 2021 - May 2022
 - Reduced **search p95 latency by 30%** by building **Python + SQL ETL pipelines**, a **MySQL-backed** inverted index, schema and query optimizations, and Flask/Docker/GCP asset services.
 - Designed **data models** with a lightweight ontology and provenance linking across **4 multilingual corpora**, enabling drill-down from aggregate lexical insights to individual source records.
 - Developed a **Flutter** Android/iOS learning application serving **3 languages** and **50,000+ vocabulary entries**, improving mobile-first access to multilingual content and teaching workflows.
 - Prototyped predictive-text and **learning-analytics** features using **Python and SQL**, increasing lesson completion and understanding by **40%**.
- **Software Developer** | *Bharat Tyres Pvt. Ltd, Anand, Gujarat, India* Sep 2020 - Sep 2021
 - Architected **Next.js** microservices on AWS EC2/ALB/RDS with **PostgreSQL**, Redis caching, and **SQL query tuning**, reducing API p95 latency by **40%** and increasing throughput by **20%**.
 - Built **Docker and Kubernetes CI/CD workflows** for automated zero-downtime deployments, improving release reliability and reducing production downtime by **30%**.
 - Implemented **JWT and OAuth 2.0** authentication and authorization across microservices, supporting user-level and service-level access control.

Projects

- **Cache Up** | *Python, FastAPI, Next.js, PostgreSQL, Pinecone, MCP, OpenAI-compatible APIs*
 - Built **Cache-Up** end to end, a full-stack tech-news summarization platform (**FastAPI + Next.js**) with a scheduled **data ingestion pipeline** indexing **100+ Hacker News posts/day** via MCP and OpenAI-compatible APIs.
 - Supported **100+ requests/second** with a FastAPI backend, Next.js/Tailwind frontend, **PostgreSQL** storage layer, Pinecone vectorized tag search, Vercel release flow, and DigitalOcean hosting.
 - Achieved **99.9% uptime** and reduced debugging time by **45%** via automated **Pytest** test suites in CI, ingestion automated every **5 minutes** on a Raspberry Pi, and real-time **Sentry monitoring**.